Storage Networking Industry Association

Assessing and Designing Storage Network Infrastructures

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Audience Poll

- **What are your storage issues?**
  - Better Storage Management
  - Improved Performance & Data Availability
  - Disaster Recovery & Backup/Restore Capabilities
  - Other ?
- **What storage infrastructure do you have or plan to have in place?**
  - Direct-attached Storage
  - SAN (Storage Area Network)
  - NAS (Network Attached Storage)
  - Integration with other protocols
- **Have you attended other SNIA Tutorials?**

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Agenda

- **Infrastructure Design Overview**
  - Components
  - Connectivity Infrastructure
  - SAN & NAS
  - Technology Building Blocks
- **Designing Storage Networks to Needs**
  - Entry-level Considerations
  - Mid-level Considerations
  - Global-level Considerations
  - Best Practices
- **Conclusions**

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Assessing and Designing Storage Network Infrastructures

Infrastructure Design Overview

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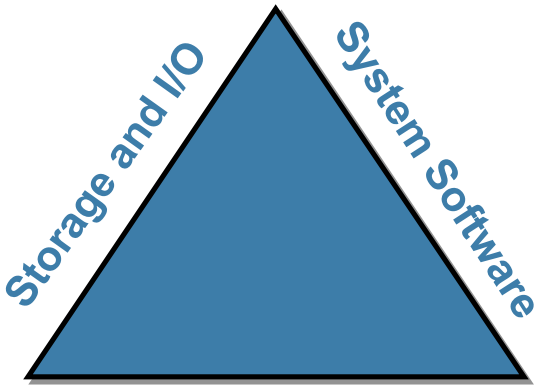
Infrastructure Design Structure

- **Connectivity Infrastructure**
  - Networks, buses
  - Physical and logical connectivity elements
- **Storage & I/O Infrastructure**
  - Devices/subsystems
  - HBAs, NICs
  - Hubs, Switches & Directors
- **System Software Infrastructure**
  - File systems
  - Volume managers
  - Management software



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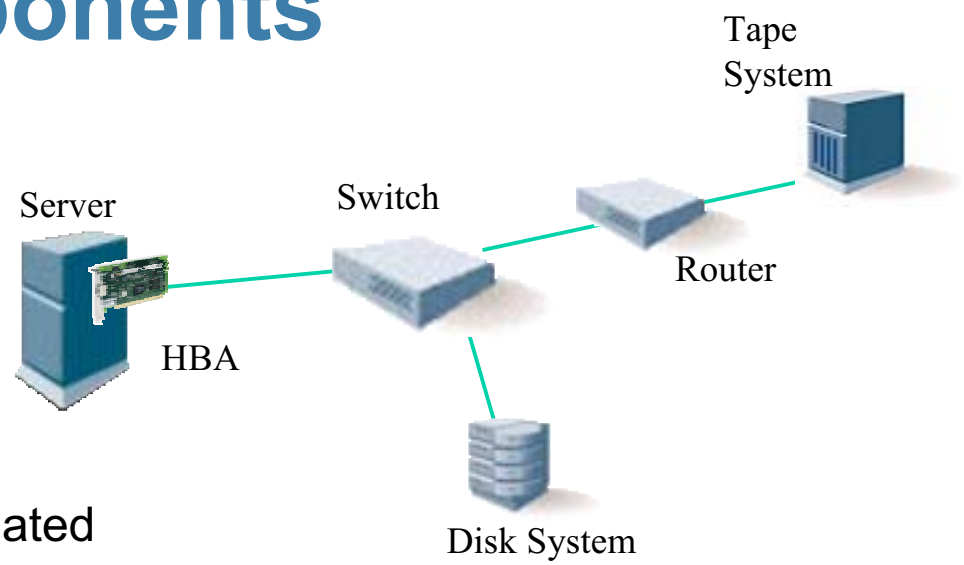
Components



- **Storage Options (Targets)**
  - Disk drives
  - JBOD
  - RAID disk subsystems
  - Solid State Disks
  - Tape/optical drives and automated tape/optical subsystems
- **Server Options (Initiators)**
  - PC Servers, UNIX Servers & Systems
  - FC Host Bus Adapters (HBAs)
- **Connectivity Options**
  - Bridges, Routers & Gateways
  - Hubs, Switches & Directors
  - Optical & Copper Cables
  - GBICs, SFPs, SFF Connectors

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Two Approaches to Network Storage

SAN

- Network connectivity between systems and storage
- Block Level Access
- Scaleable Performance for on-demand storage
- Increased Data Integrity
- Extended Distance
- Consolidate Storage Resources for effective centralized management

NAS

- Client/server access to files on a specialized file server
- File Level Access
- Peer to peer communications
- Clients redirect file requests over the network
- Servers maintain file storage system & access control

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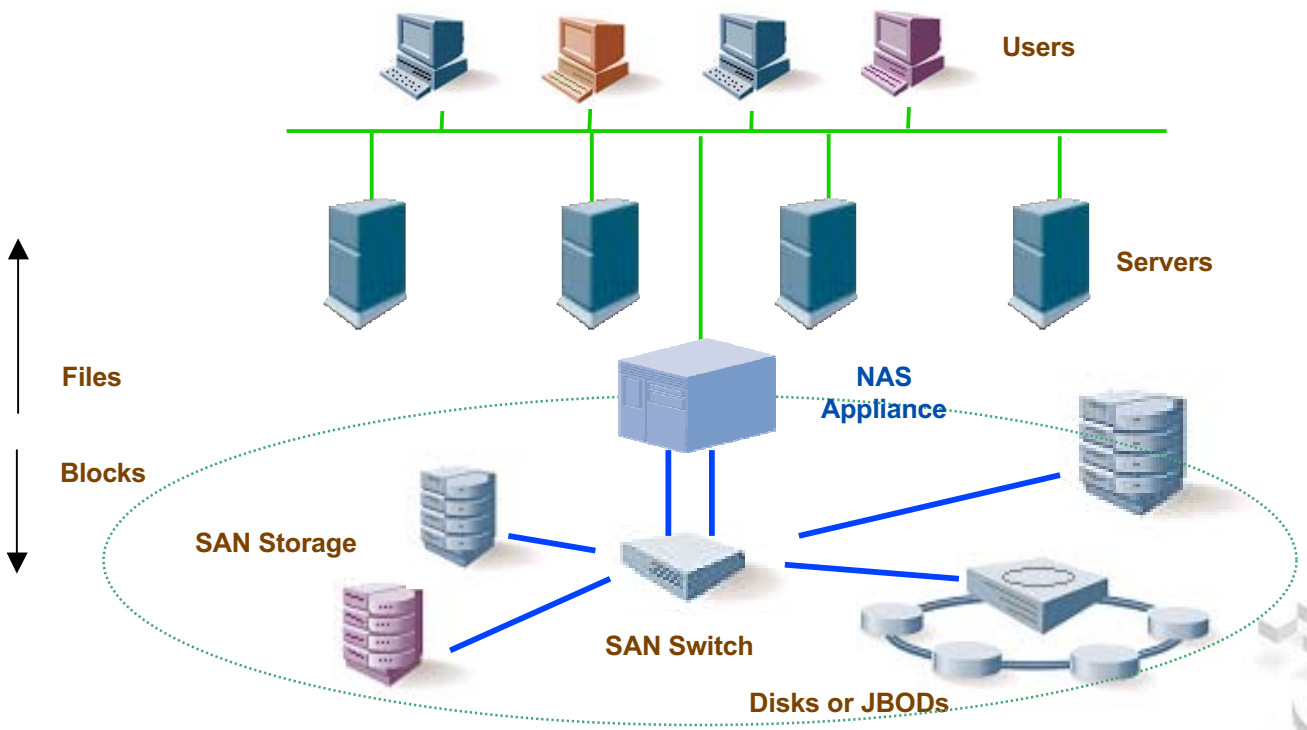
SANs as Infrastructure for NAS

- NAS ‘front end’ & SAN ‘back end’
- Two technologies co-exist (not an either/or)



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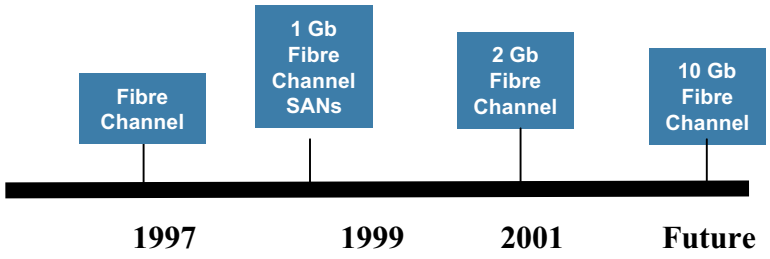
Assessing and Designing Storage Network Infrastructures

Technology Building Blocks

Why Fibre Channel?



- Established in late 1980s, 2Gb last year & 10Gb on roadmap
- $1.3B Fibre Channel SAN Component Market in 2000 & growing to $10.8B by 2005 (53% CAGR)*
- 2Gb FC doubles performance at the same price as 1Gb/s Ethernet
- 2Gb FC is plug-compatible with 1Gb FC (devices auto negotiate)
- Applications are driving higher data rates (i.e. Video)
- Read/Write Operations on 2Gb FC HBAs show dramatic improvements with 98.5% real utilization on saturated lines
- Serial SCSI; FCP protocol
- Minimal error rates for network technology
- Credit-based flow control

* Gartner/DataQuest 6/2001 Fibre Channel SAN Components

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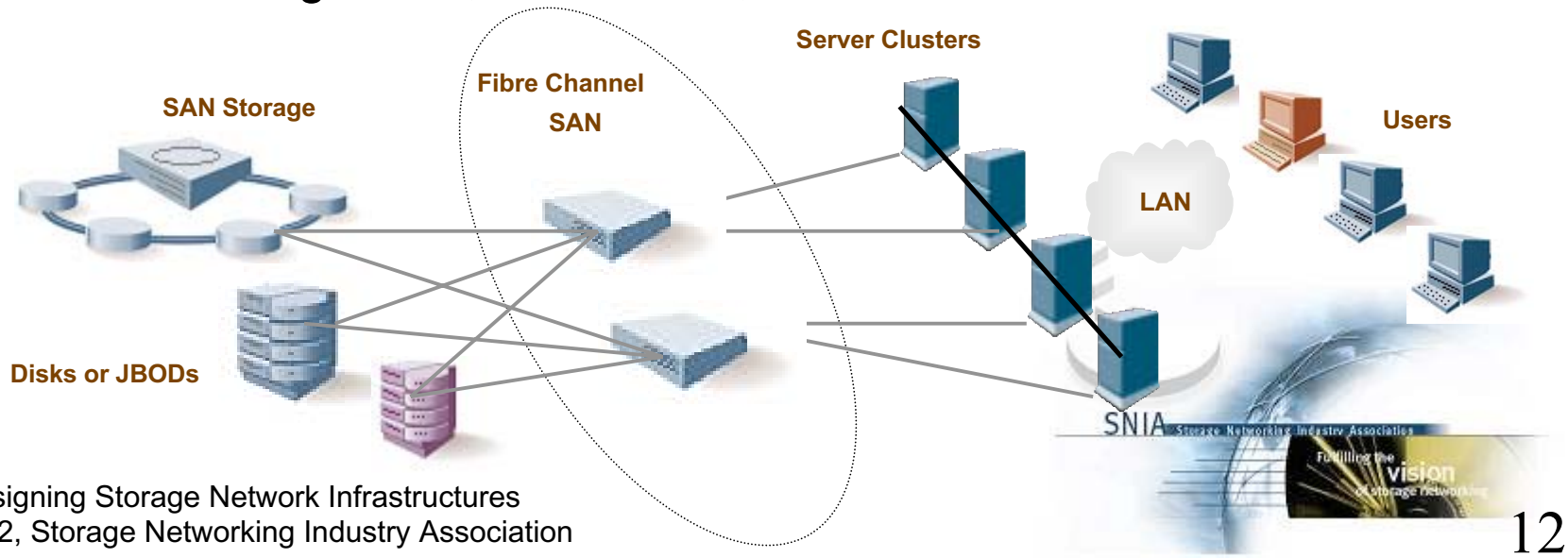
Why a SAN?

- **Scaleable Performance**
  - SAN Infrastructure expands easily for on-demand storage
- **Increased Data Integrity**
  - Dual-paths for redundancy, fault tolerance & performance
- **Extended Distance**
  - Data access from anywhere at anytime
- **Consolidate Storage Resources**
  - Easier management, centralized, and more cost effective



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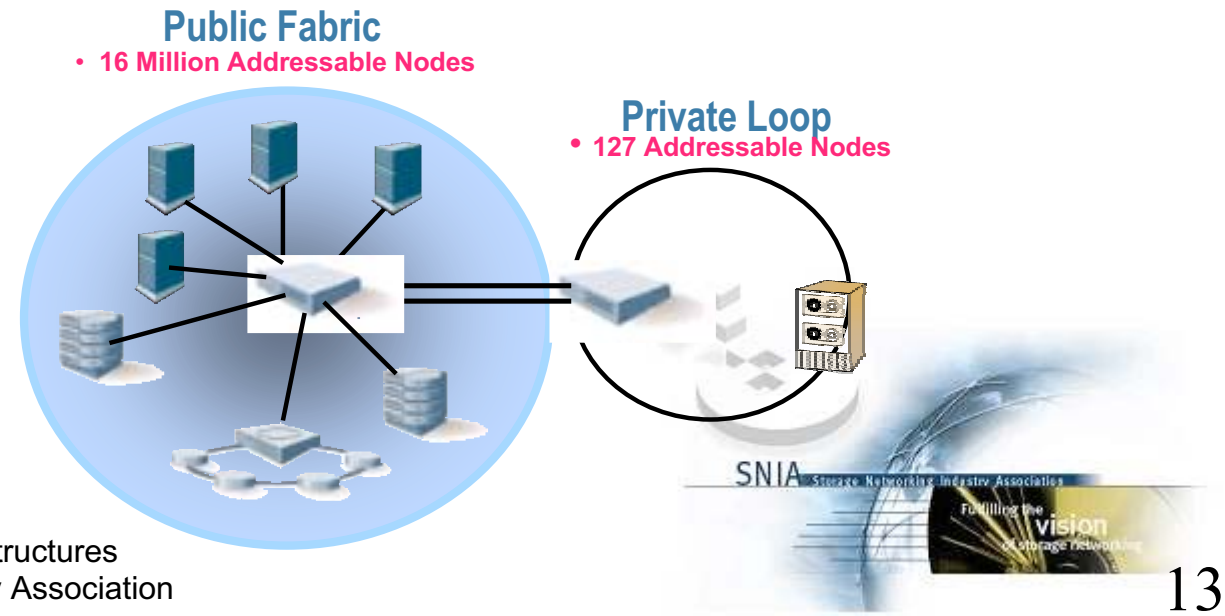
Fibre Channel Topology Selection

- **Private loop Interfaces** (8-bit addressing)
- **Fabric / Public loop Interfaces** (24-bit addressing)
  - Public loop nodes can login and access fabrics
- **Combining Loops and Fabrics**
  - Isolate Secure Data by not allowing fabric access
  - Multiple loops in a fabric increase the number of available ports



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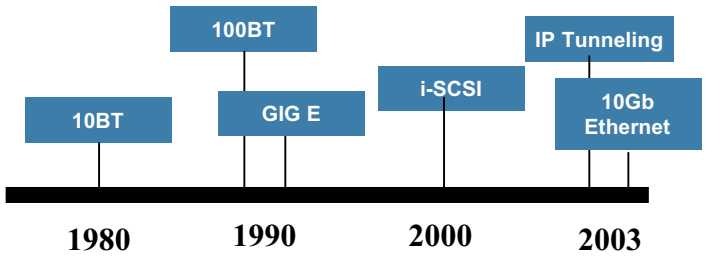
Why iSCSI ?

Serial SCSI over TCP/IP networks



- **IETF standards work**
- **IP Routing**
- **Ethernet assumptions used**
  - Gigabit Ethernet+ implied
  - Similar in scope to FC link credits
- **iSCSI uses TCP-session flow control**
  - TCP algorithms consume CPU cycles at Gb speeds
  - TCP acceleration technology (TOE) started shipping
- **Leverages existing Ethernet/TCP/IP Technologies and Know-How in the market**

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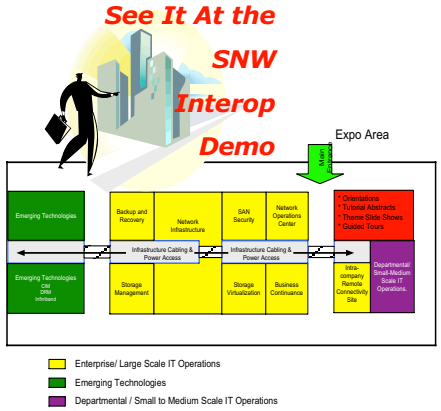
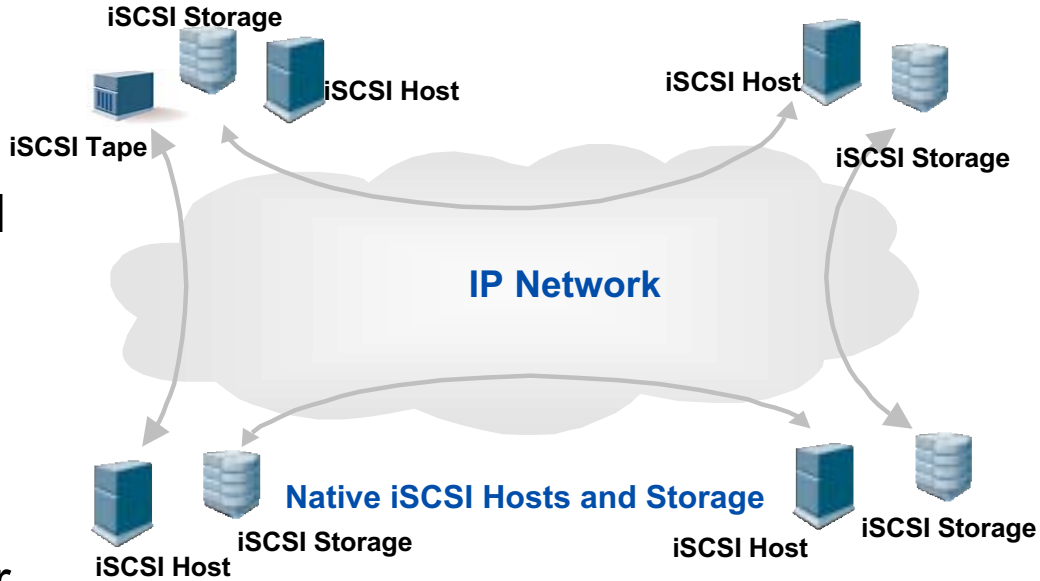
iSCSI SANs



- Adoption at the low-end of the market first
- Do you use the existing IP network?
- Trained IP networking staff and support? (What's a LUN?)

- Do you build an independent network for storage I/O on IP or Fibre Channel?
- Long-term infrastructure requirements?



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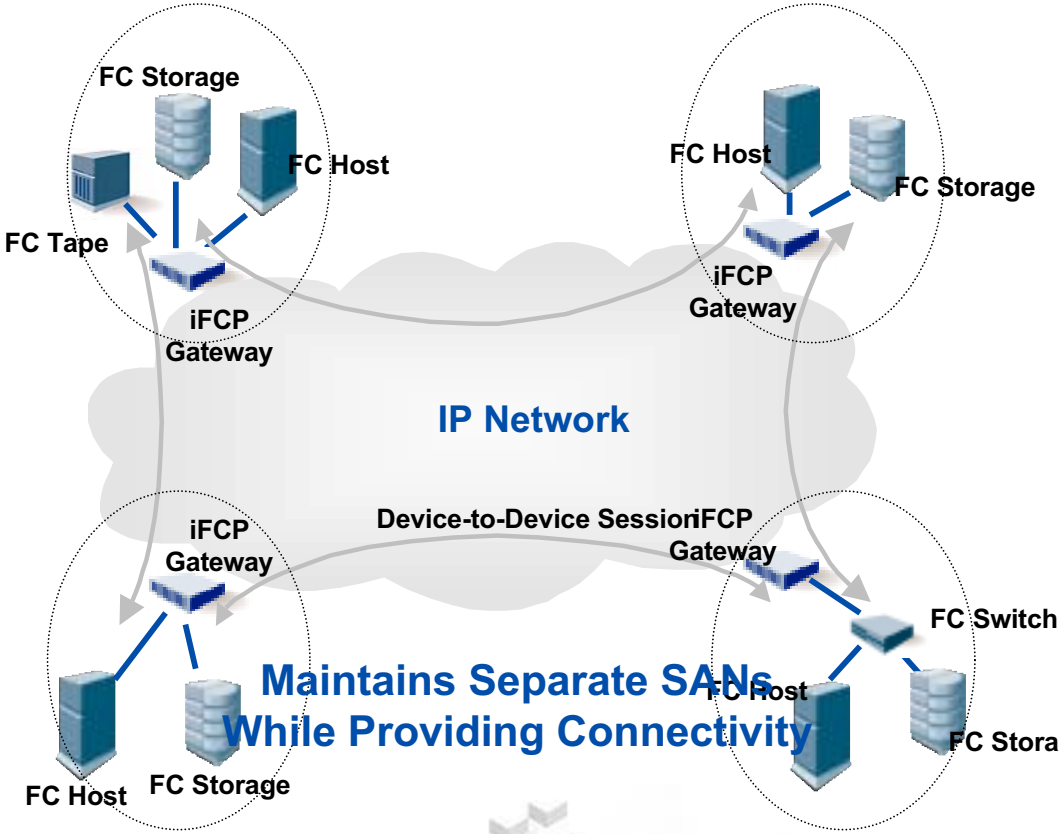
iFCP SANs

- Migration from FC to IP SANs
- Standards track within IETF
- FC cabling and services to FC end nodes
- Building SAN infrastructure with Gigabit Ethernet switches and FC end devices
- iSNS (Internet Simple Name Server) is the corresponding naming service
- FC E-port compatibility



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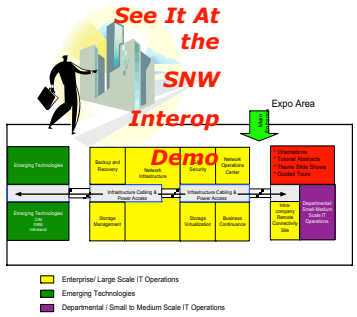
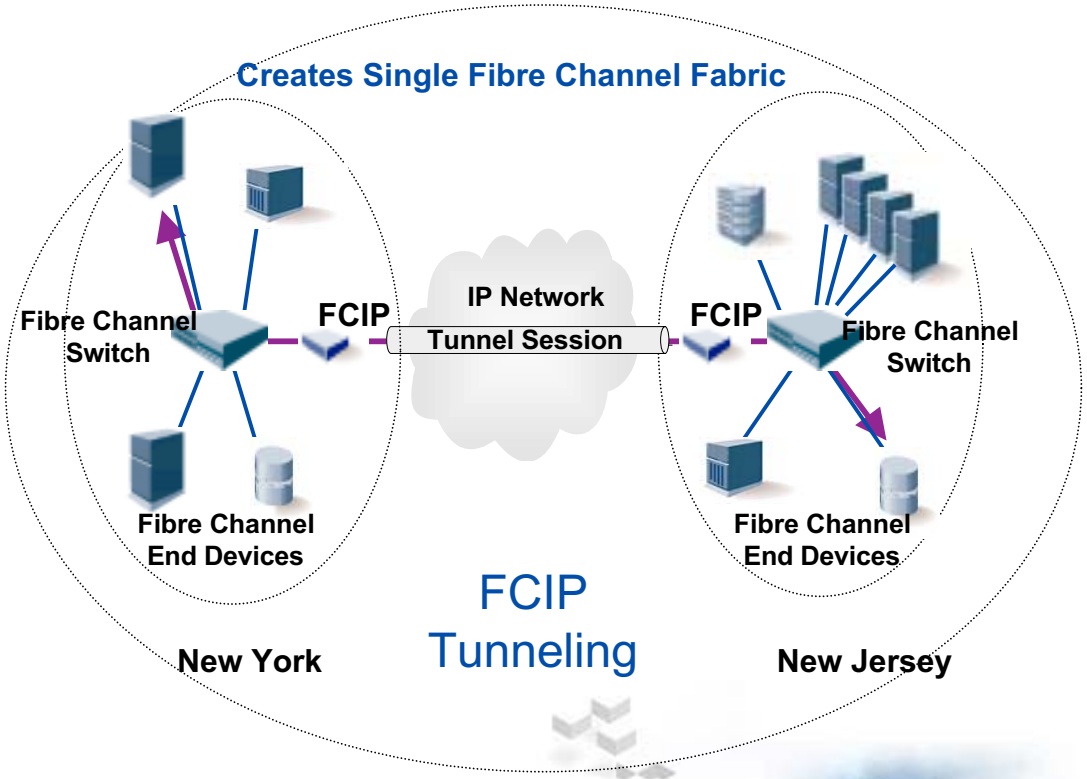
Connecting SANs with FCIP



- **Fibre Channel Extension**
  - Point-to-point IP tunneling
  - Wrap entire FC frame in TCP/IP for transmission
  - Strip TCP/IP at receiving end
  - On standards track within IETF IP Storage Work Group
- **Leverage IP for Long Haul Applications**
  - Asynchronous mirroring / backup / remote vaulting
  - WAN bandwidth may limit type of storage applications that can be supported



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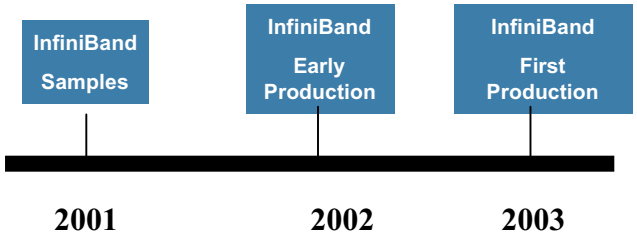
Why InfiniBand™?



- **Designed for server clusters and I/O for storage networks**
  - A shift to a switched fabric based I/O architecture
  - Connect to the I/O Fabric with one or more HCAs (Host Channel Adapters)
  - Devices attach to the fabric via TCA (Target Channel Adapters)
- **Scalable Performance**
  - De-couples the I/O subsystem from memory
  - Utilizing channel based point-to-point connections vs. shared bus load and store configuration
  - 2.5 Gbit/sec wire speed connection with one, four or twelve wire link widths
  - Offers scalable performance through multi-link connections and a host of interoperable link speeds with range of 500Mbyte/s to 6 Gbyte/s
  - Aggregates system performance more than wire speed
  - Supports thousands of nodes and multiple links and link widths
- **Availability**
  - Redundant Paths and Fabrics
  - Isolated Fault Zones
  - Layered Error Management
- **For more info: www.infinibandta.org**

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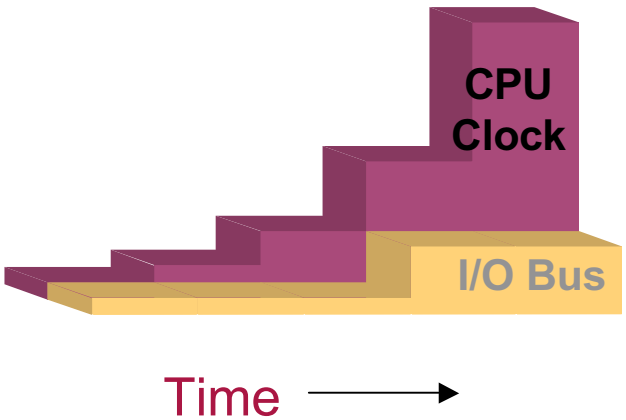
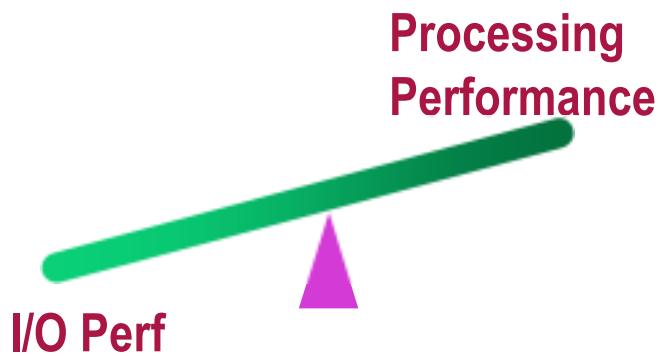
Why I/O Matters





- CPU performance doubles every 12-18 months according to Moore's Law
- I/O Bus bandwidth doubles about every three years

- A new I/O technology is needed to bring balance to server system performance

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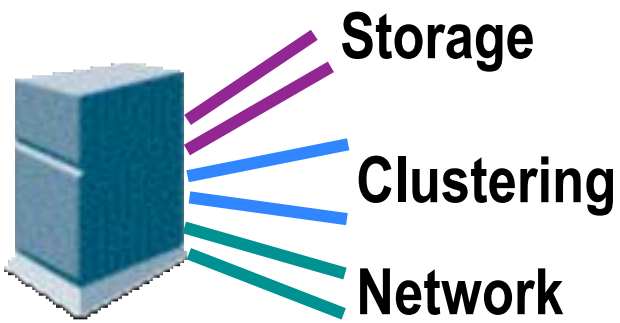
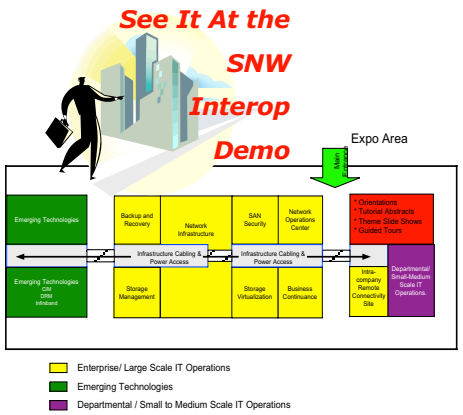
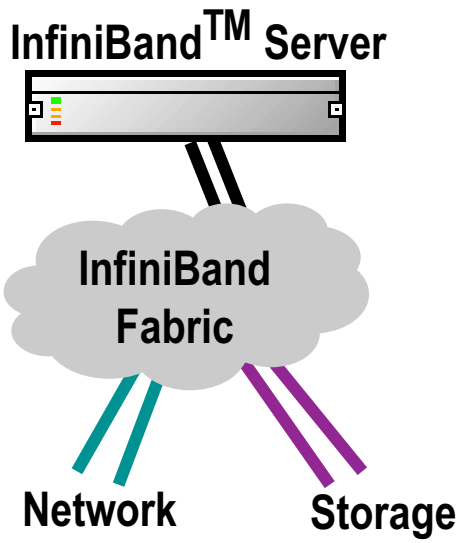
I/O Sharing

Fabric Connectivity







- Independent Scaling of Processing & I/O
- Decreased Cabling Requirements
- Increased Data Center Flexibility & Performance
- Servers become easier to deploy & manage

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Storage Area Networks (SAN)

Designing Storage Networks to Needs

High-Availability SANs
Direct Costs of Downtime can be High

Industry	Average Loss/Hr.
• Brokerage Operations	$6,450,000
• Credit Card Authorizations	$2,600,000
• E-commerce	$ 240,000
• Package Shipping Services	$ 150,250
• Home Shopping Channels	$ 113,750
• Catalog Sales Center	$ 90,000
• Airline Reservation Center	$ 89,500
• Cellular Service Activation	$ 41,000
• ATM Service Fees	$ 14,500

Source: Gartner Group

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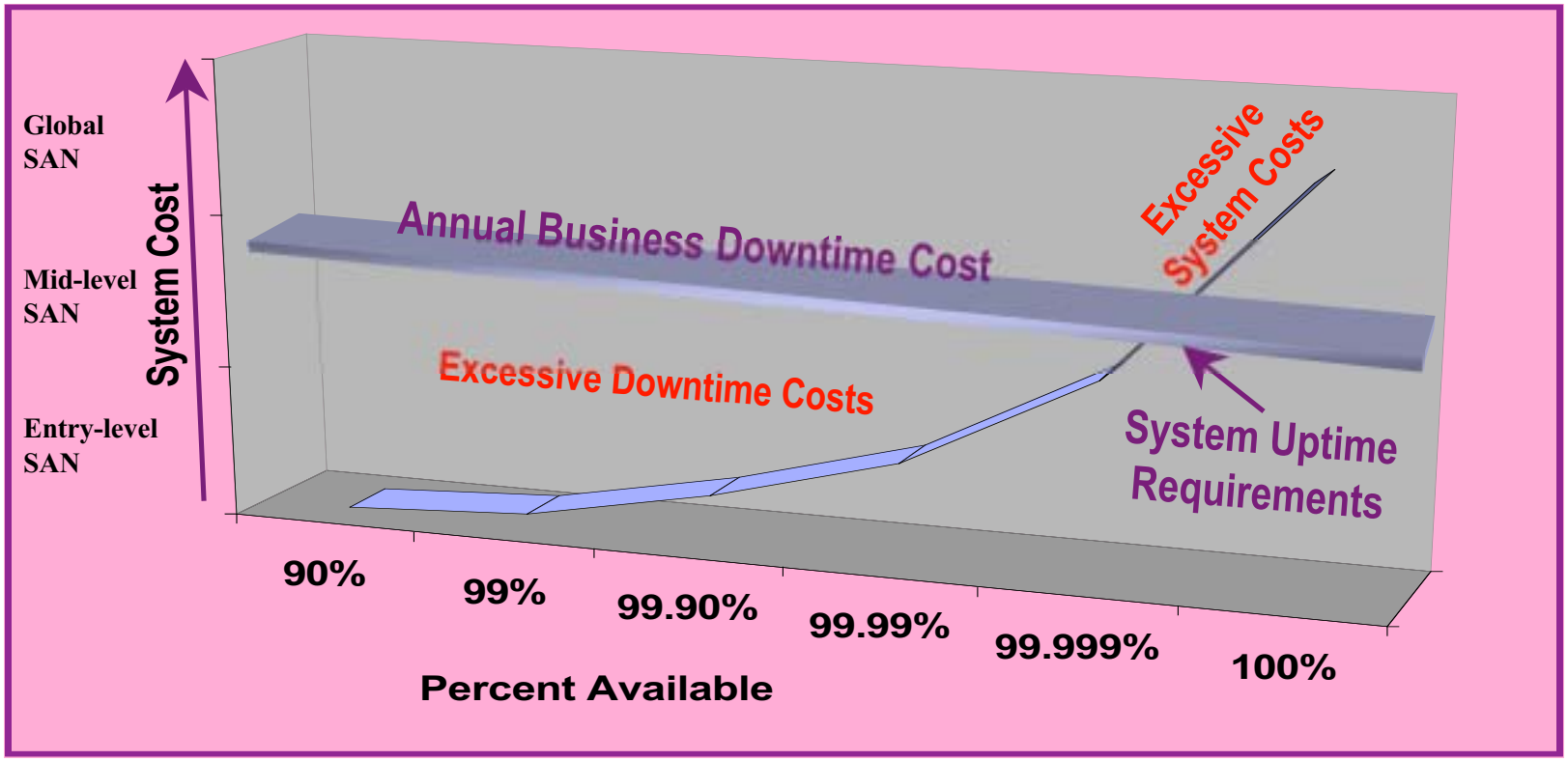
Cost vs. Uptime Requirements



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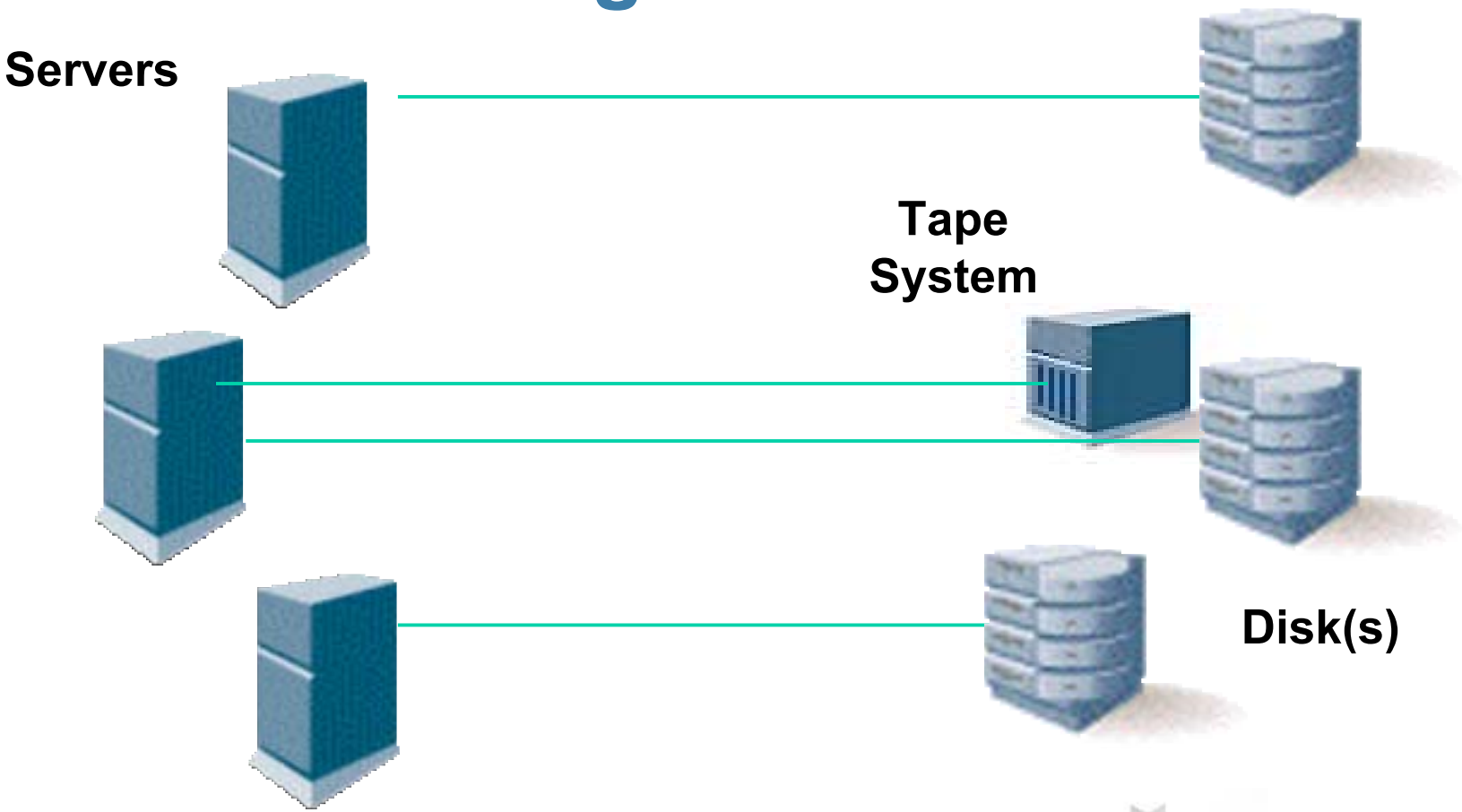
Storage Area Networks (SAN)

Entry-level SANs

Moving from DAS…



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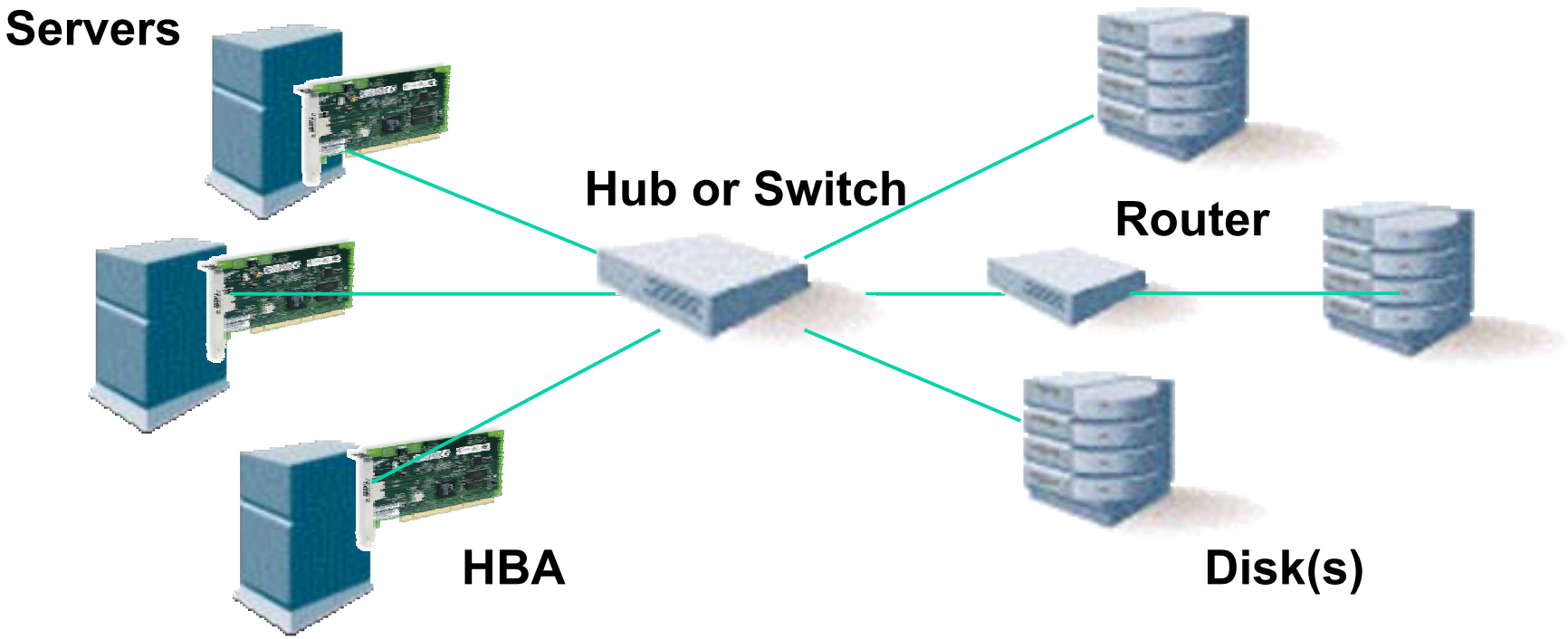
…to SAN



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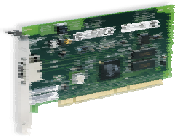
Review: New Components for SAN

- **Host Bus Adapters (HBA)**
  - Attaches initiators (servers) to SAN Fabric
  - More intelligent than NIC by offloading I/O minimizes CPU utilization
- **Hubs**
  - Minimal speed, connections & intelligence
  - Provides star topology, shared bandwidth, interrupts operations (LIPs)
- **Bridge or Router**
  - Bridges multiple technologies (ie: FC to SCSI)
- **Fabric Switches**
  - Provides intelligence fabric
  - SAN scalability to millions of devices
- **Directors**
  - Highly redundant providing 99.999% availability
- **Cables, GBICs, SFPs & SFF Connectors**



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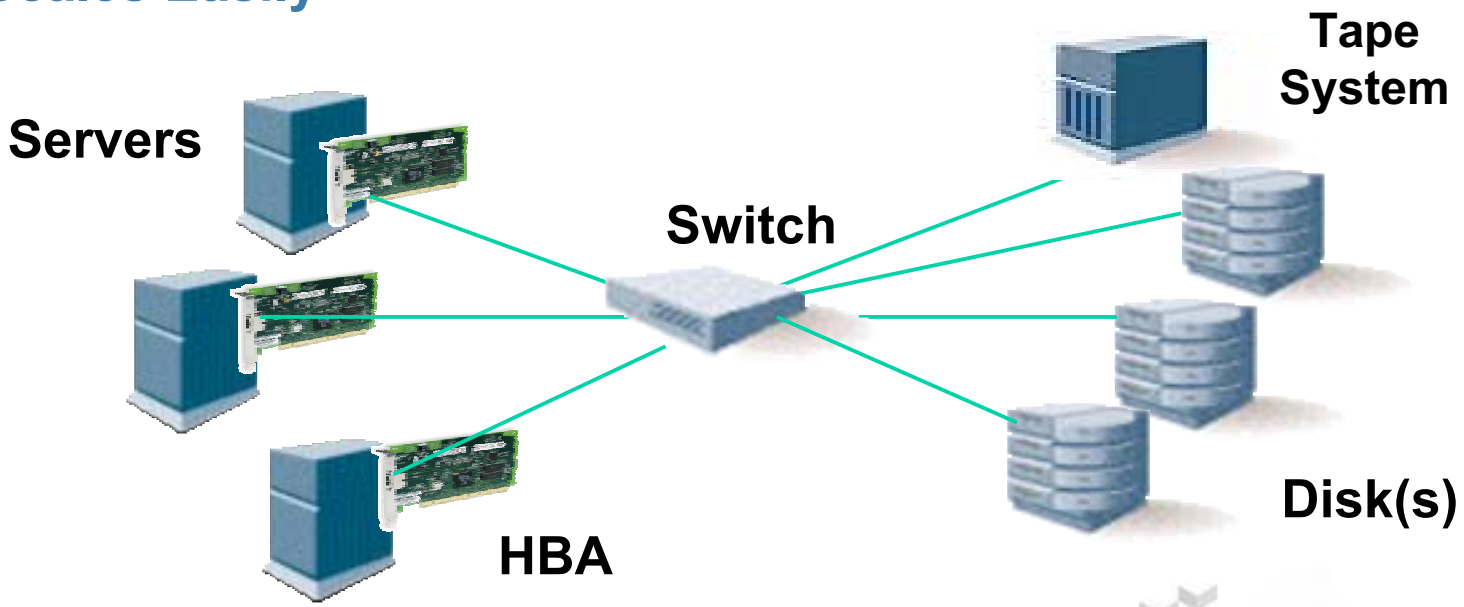
SAN Backup/Restore

- Increase Performance
- Device Sharing
- Scales Easily



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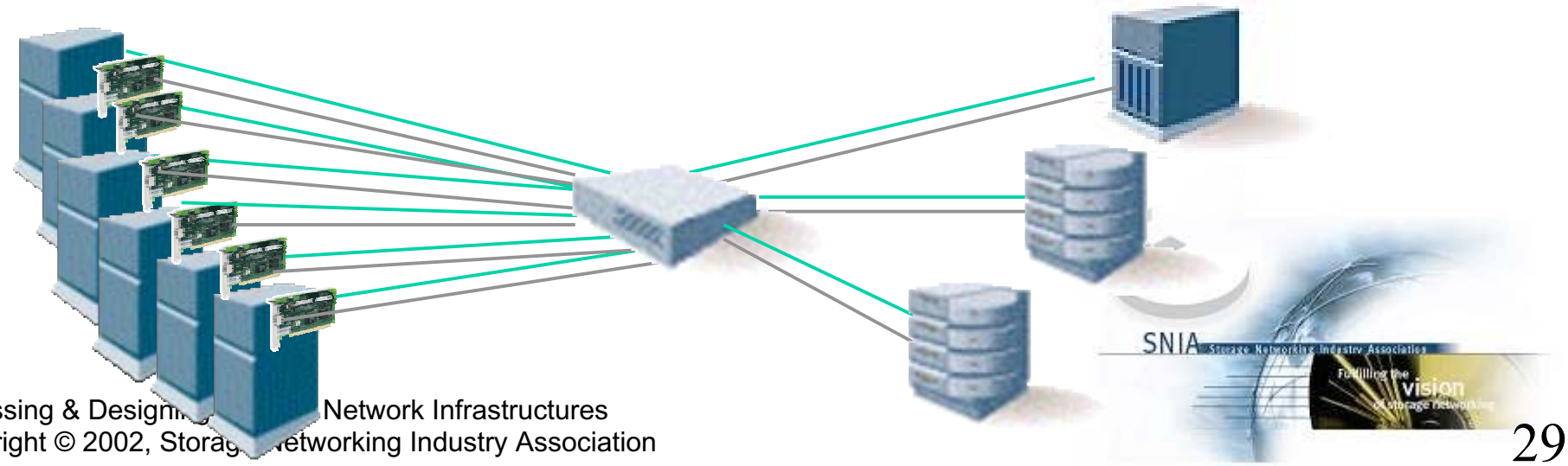
Backup Application Example

Business IT Problem:

- Rapid storage growth
- New Applications being deployed
- Consolidate 10 Servers
- Shrinking backup window
- Future-Proof the Infrastructure

Business IT Solution:

- Consolidate existing storage in SAN
- Consolidate servers in SAN
- Add new storage for planned growth
- Add tape backup system
- Add backup management application



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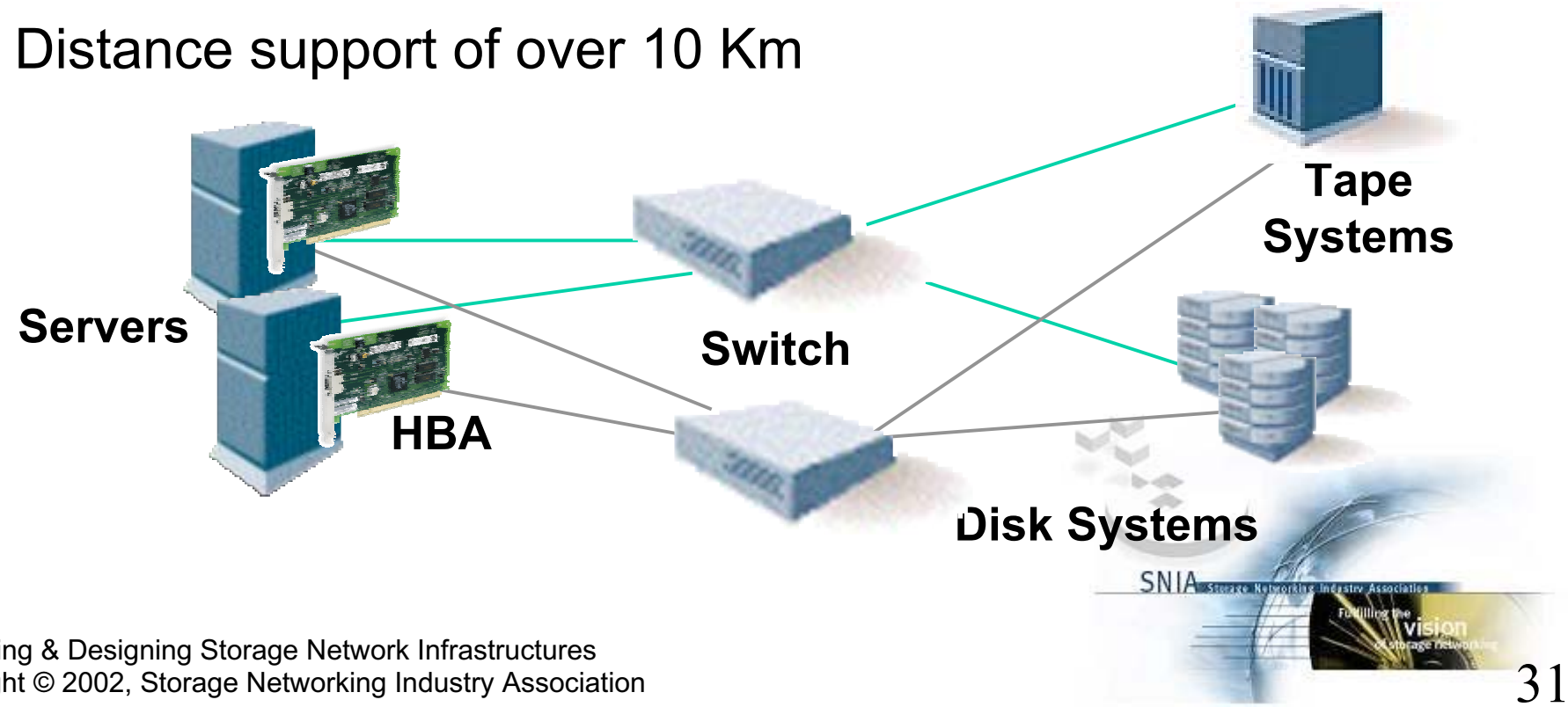
Storage Area Networks (SAN)

Mid-level SANs

SANs for High Availability

- Redundant data paths
  - All storage and servers have at least two interfaces
- Performance that scales
  - Each FC 1Gb connection adds 100 MB/s throughput in each direction (or 200MB/s) and with 2Gb double it!
- Distance support of over 10 Km



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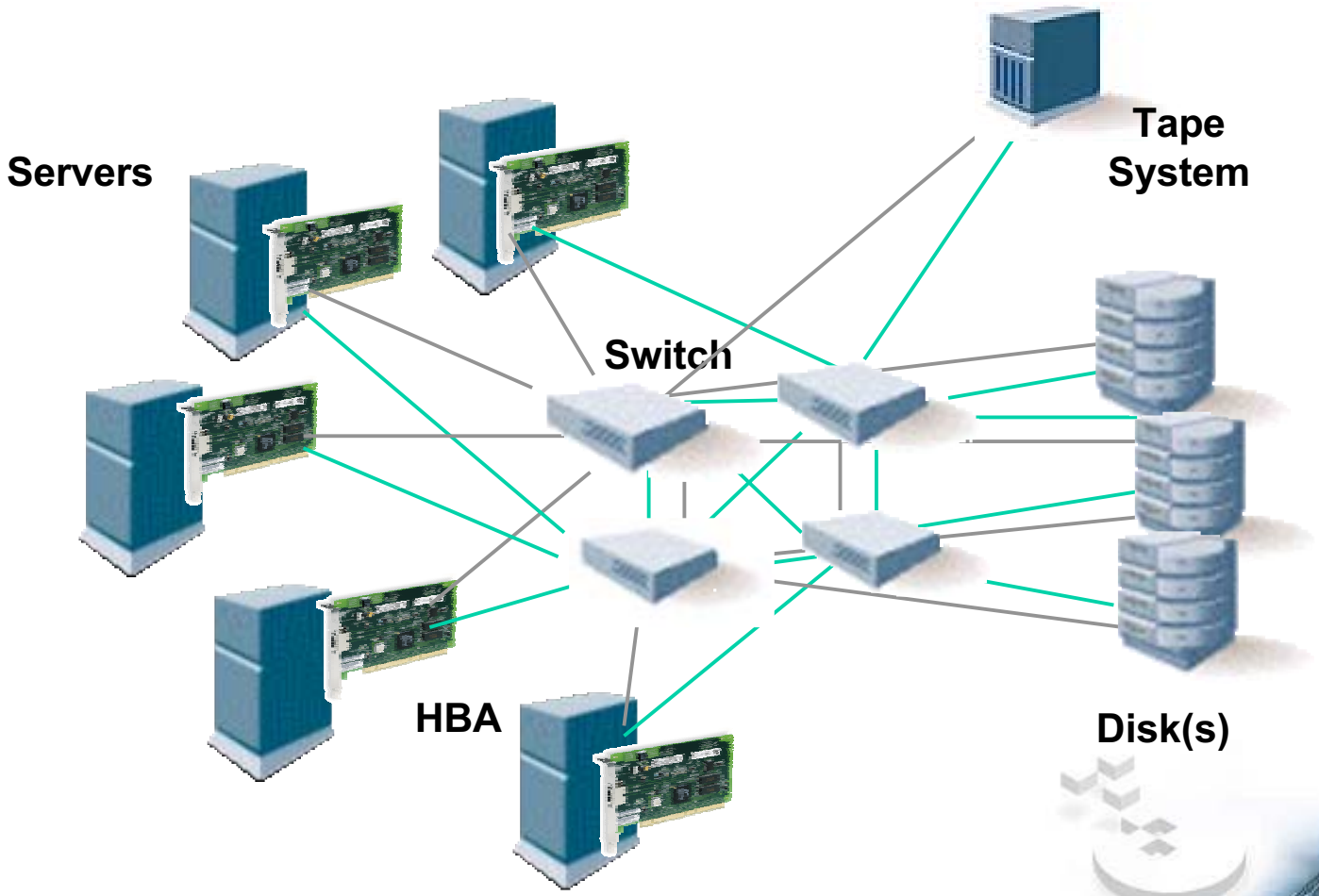
SAN: Dynamic Scalability



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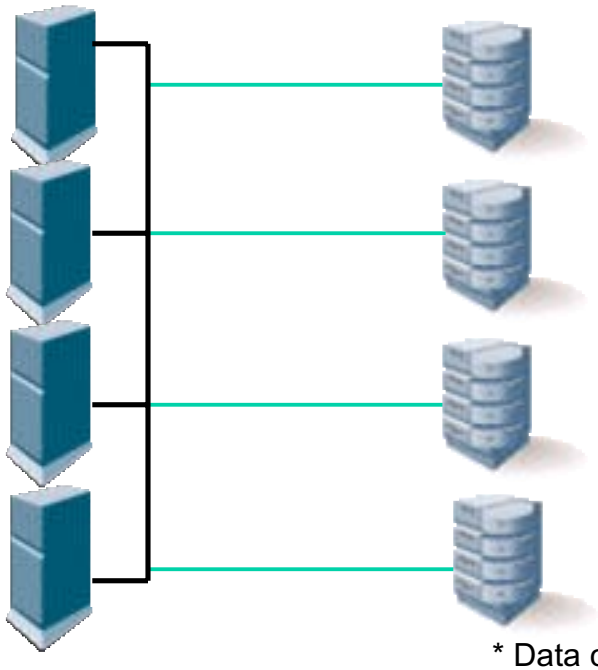
SAN Storage Consolidation

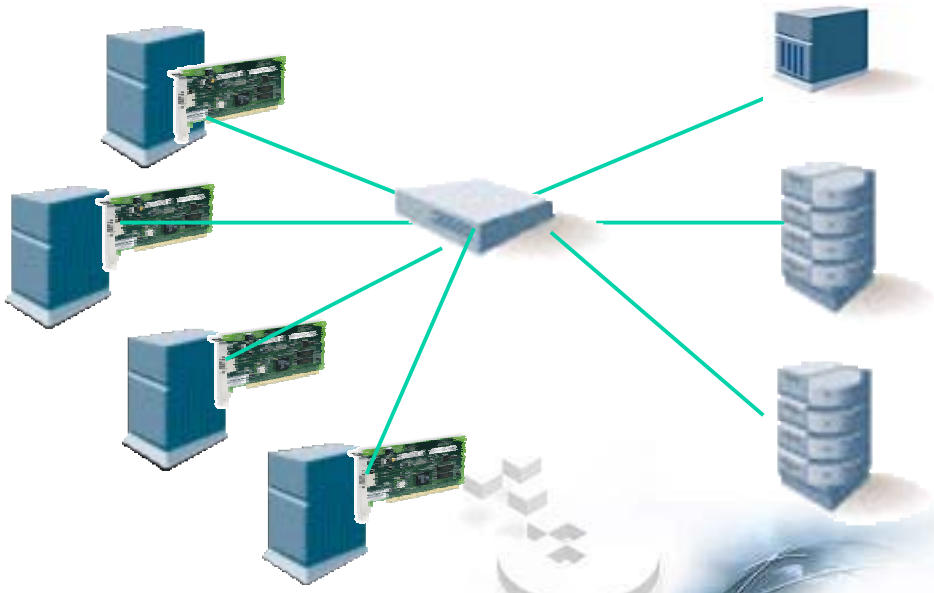
Multiple Servers use a single pool of partitioned disk

- Easier to manage (storage is centralized)
- Less free disk space required

500 Gb @ 50% Utilization per system or 2TB with 1TB Free*



1.5 TB w/.5 TB Free = Save cost of 500 Gb*



* Data could vary depending on configuration

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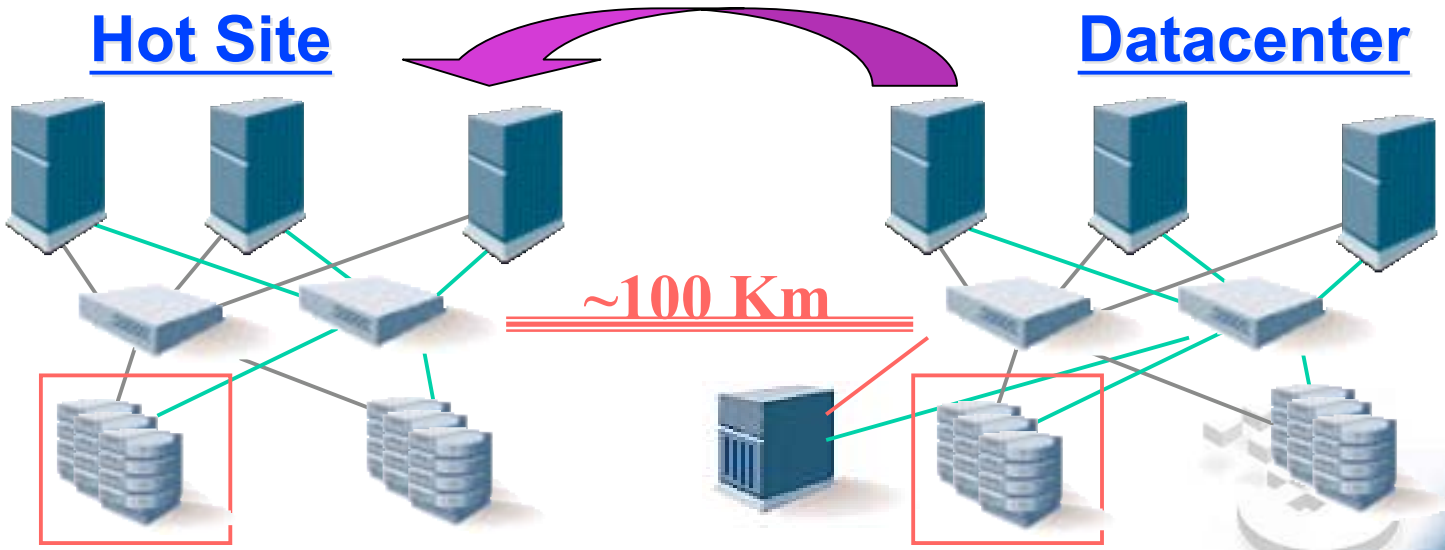
Metro Site Infrastructure

- Higher level of DR Protection
- Synchronous mirrored applications over distance
- Off-site backups
- Auto-failover if disruption is detected
- High availability environment



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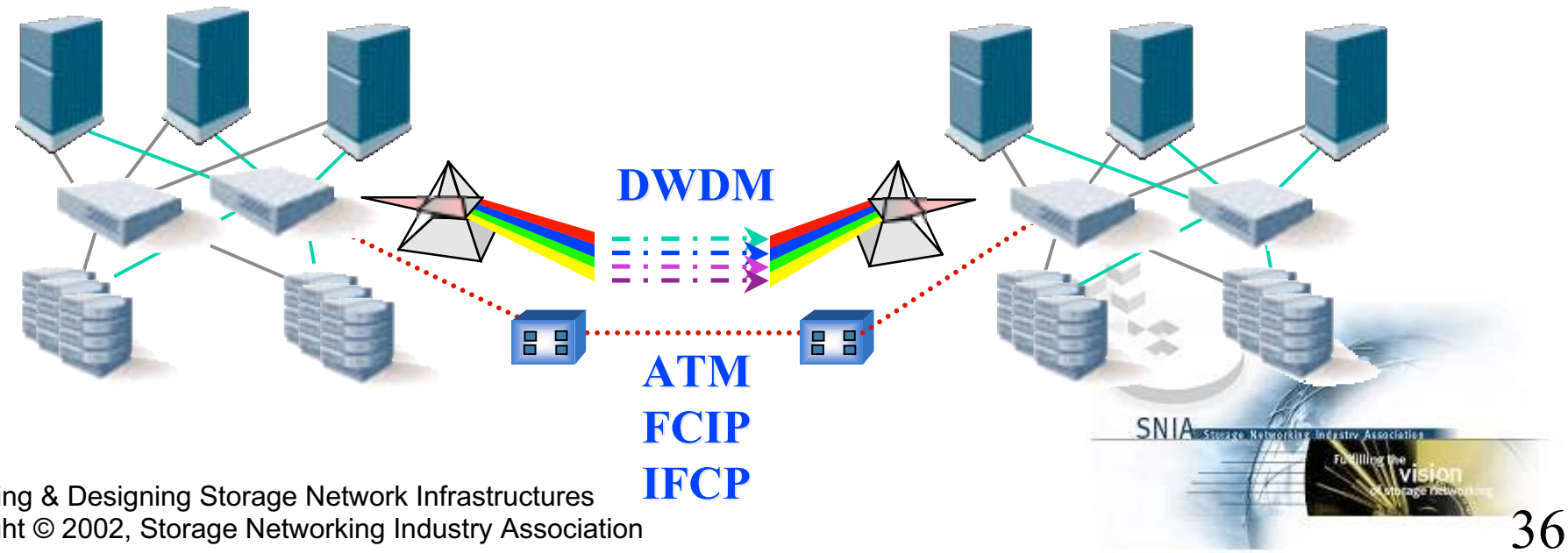
Storage Area Networks (SAN)

Global SANs

Fabrics for Greater Distance: MAN's & WAN's

- **Metro Area Networks (MANs)**
  - Distances of 100+ Km
  - Redundant paths are possible
  - Scalable performance
  - Applications
    - Backup and Archival – Async.
    - Disk Mirroring (Sync)
- **Wide Area Networks (WANs)**
  - Distances of 5,000 Km
  - QoS options
  - Costs can range from $1,000 to $20,000/month/connection
  - Performance limited:155 Mb/sec, 622 Mb/sec, etc.
  - Applications
    - Disk Mirroring (Async)
    - Backup and Archival



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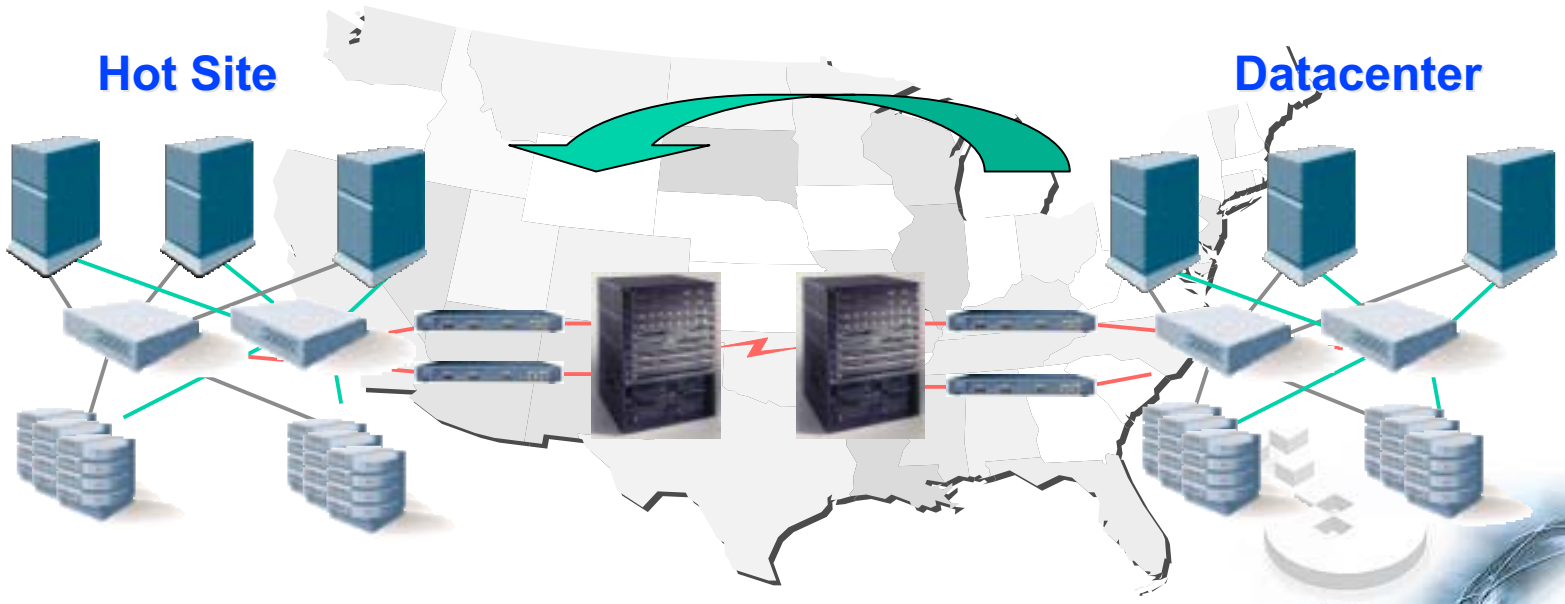
Global Site Infrastructure

- Protection from largest disasters
  - Exposures: Hackers
- Asynchronous mirrored applications over distance
- Off-site backups
- Auto-failover if disruption is detected



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Closing Thoughts

- Pilot Projects Required
- Plan for growth with scalable solutions
- Incorporate existing infrastructure
- Plan and practice disaster recovery
- Increase productivity with good management applications & practices
- Reduce downtime and minimize impact for upgrades
- Create a safe network environment

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Q&A / Feedback

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